Qualification Pack

Unmanned Aerial System

QP Code: ELE/Q6707

Version: 1.0

NSQF Level: 4.5

Electronics Sector Skills Council of India || 155, 2nd Floor, ESC House Okhla Industrial Area-Phase 3
New Delhi- 110020 || email:anu@essc-india.org

Contents

ELE/Q6707: Unmanned Aerial System

Brief Job Description

Unmanned Aerial System (UAS) training program equips learners with advanced skills in drone-based sensing, imaging, data processing, and autonomous operations. The program covers integration and use of advanced sensors such as LiDAR, infrared, and 3D cameras, along with photography, videography, and geospatial data collection. It introduces core autonomy concepts including filtering techniques, PID control, and SLAM, and enables practical application in 3D modelling, simulation, thermography, and industrial asset inspections, preparing participants for advanced, data-driven UAS applications

Personal Attributes

Unmanned Aerial System (UAS) training program equips learners with advanced skills in drone-based sensing, imaging, data processing, and autonomous operations.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. ELE/N6731: Work organization and management (Unmanned Aerial System)
2. ELE/N6732: Communication and interpersonal skills (Unmanned Aerial System)
3. ELE/N6733: Assemble/repair, and maintain an unmanned aerial system (UAS/RPAS) (Unmanned Aerial System)
4. ELE/N6734: Setup, program, and operate the UAS (Unmanned Aerial System)
5. ELE/N6735: Manual flight demonstration and emergency procedures (Unmanned Aerial System)
6. ELE/N6736: Autonomous flight planning and demonstration (Unmanned Aerial System)
7. ELE/N6737: UAS Advanced piloting and embedded vision (Unmanned Aerial System)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	E-Mobility and Battery
Occupation	Product Design & Development-EM&B
Country	India

Qualification Pack

NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8212.0400
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	06/02/2028
NSQC Approval Date	06/02/2026
Version	1.0
Reference code on NQR	QG-4.5-EH-05200-2026-V1-ESSCI
NQR Version	1

Remarks:

NA

ELE/N6731: Work organization and management (Unmanned Aerial System)

Description

This NOS is about to defines the competencies required to plan, coordinate, and manage UAS operations efficiently while ensuring regulatory compliance, safety standards, and effective resource utilization.

Scope

The scope covers the following :

- Work organization and management (Unmanned Aerial System)

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Maintain a healthy, safe, and secure working environment by following organizational safety policies, statutory requirements, and standard operating procedures.

PC2. Apply organizational safety procedures related to electrical safety, safe handling of tools, batteries, and hazardous materials, including fire safety practices.

PC3. Select and use appropriate Personal Protective Equipment (PPE) according to task requirements and work environment conditions.

PC4. Plan, prioritize, and re-prioritize work activities using effective work planning and scheduling techniques to meet timelines, productivity targets, and quality requirements.

PC5. Execute assigned tasks in compliance with defined quality standards and stages of the quality assurance process, and explain their importance to team members.

PC6. Demonstrate effective team working by giving and receiving constructive feedback, supporting colleagues, and promoting workplace ethics and etiquette.

PC7. Identify, select, and apply appropriate types of UASs in line with their applications, applicable rules and regulations, and essential safety precautions.

PC8. Follow organizational reporting standards, waste management and recycling practices, and continuously update professional knowledge in line with industry trends and developments.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Comprehensive knowledge of organizational safety policies, statutory regulations, standard operating procedures (SOPs), and safe work practices to maintain a healthy, safe, and secure working environment.

KU2. Understanding of electrical safety standards, safe handling of tools, batteries, hazardous materials, fire prevention methods, and emergency response procedures.

KU3. Knowledge of types, selection criteria, correct usage, limitations, and maintenance of PPE based on task requirements and workplace hazards.

KU4. Understanding of work planning techniques, scheduling methods, prioritization strategies, quality assurance stages, documentation requirements, and productivity benchmarks.

KU5. Knowledge of different types of Unmanned Aerial Systems (UAS), their applications, applicable rules and regulations, safety precautions, reporting standards, waste management practices, and the importance of continuous professional development.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to apply organizational safety procedures, follow statutory requirements, implement electrical and fire safety measures, and maintain hazard-free work conditions.

GS2. Skill in selecting, inspecting, and properly using appropriate PPE, tools, batteries, and hazardous materials while adhering to safety protocols.

GS3. Proficiency in planning, prioritizing, and executing tasks effectively to meet deadlines, productivity targets, and quality standards.

GS4. Ability to perform tasks in accordance with defined quality standards, follow reporting procedures, manage waste responsibly, and document activities accurately.

GS5. Competence in collaborating with team members, giving and receiving constructive feedback, maintaining workplace ethics and etiquette, complying with UAS regulations, and continuously upgrading professional knowledge.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	8	-	-
PC1. Maintain a healthy, safe, and secure working environment by following organizational safety policies, statutory requirements, and standard operating procedures.	-	1	-	-
PC2. Apply organizational safety procedures related to electrical safety, safe handling of tools, batteries, and hazardous materials, including fire safety practices.	-	1	-	-
PC3. Select and use appropriate Personal Protective Equipment (PPE) according to task requirements and work environment conditions.	-	1	-	-
PC4. Plan, prioritize, and re-prioritize work activities using effective work planning and scheduling techniques to meet timelines, productivity targets, and quality requirements.	-	1	-	-
PC5. Execute assigned tasks in compliance with defined quality standards and stages of the quality assurance process, and explain their importance to team members.	-	1	-	-
PC6. Demonstrate effective team working by giving and receiving constructive feedback, supporting colleagues, and promoting workplace ethics and etiquette.	-	1	-	-
PC7. Identify, select, and apply appropriate types of UASs in line with their applications, applicable rules and regulations, and essential safety precautions.	-	1	-	-
PC8. Follow organizational reporting standards, waste management and recycling practices, and continuously update professional knowledge in line with industry trends and developments.	-	1	-	-
NOS Total	-	8	-	-

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6731
NOS Name	Work organization and management (Unmanned Aerial System)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6732: Communication and interpersonal skills (Unmanned Aerial System)

Description

This NOS defines the competencies required to communicate effectively with team members, clients, and regulatory authorities while maintaining professionalism, coordination, and clear information exchange during UAS operations.

Scope

The scope covers the following :

- Communication and interpersonal skills (Unmanned Aerial System)

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Interpret the principles and requirements of flight planning, including identification of potential flight hazards and coordination of airspace requirements.

PC2. Identify and select appropriate types of UASs based on mission objectives, client requirements, and application-specific constraints.

PC3. Collect, analyze, and document client requirements, operational issues, and mission expectations using structured communication methods.

PC4. Read, interpret, and extract relevant technical data, specifications, and instructions from electronic documentation and publications in various formats.

PC5. Communicate technical information clearly and effectively through oral, written, and digital means while using appropriate technical terminology and professional language.

PC6. Exchange information, instructions, and advice with team members and stakeholders, seek clarifications, and provide constructive feedback using standard communication technologies.

PC7. Conduct pre-mission planning meetings with clients and internal teams to confirm mission scope, safety requirements, timelines, and documentation.

PC8. Create, update, organize, and maintain design and technical documentation such as schematics, wiring diagrams, bills of material, and knowledge-base or training materials.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Comprehensive knowledge of flight planning procedures, identification of potential flight hazards, risk assessment methods, airspace classification, regulatory permissions, and coordination requirements for safe UAS operatio

KU2. Understanding of different types of Unmanned Aerial Systems (UAS), their configurations, payload capacities, endurance, control systems, and suitability based on mission objectives and operational constraints.

KU3. Knowledge of structured communication techniques for gathering, analyzing, and documenting client requirements, operational challenges, and mission expectations.

KU4. Understanding of how to read and interpret electronic documentation, technical publications, schematics, wiring diagrams, bills of material (BOM), and system specifications in multiple formats.

KU5. Knowledge of technical terminology, documentation standards, digital communication tools, reporting protocols, and methods for maintaining organized and up-to-date technical records.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to interpret flight planning requirements, identify hazards, coordinate airspace usage, and ensure mission compliance with safety and regulatory standards.

GS2. Skill in selecting appropriate UAS platforms based on mission scope, client needs, technical constraints, and environmental considerations.

GS3. Proficiency in collecting, analyzing, and documenting client inputs and operational requirements using structured and systematic approaches.

GS4. Competence in reading technical data, extracting relevant specifications, preparing schematics, updating documentation, and maintaining organized technical records.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**8**	**-**	**-**
PC1. Interpret the principles and requirements of flight planning, including identification of potential flight hazards and coordination of airspace requirements.	-	1	-	-
PC2. Identify and select appropriate types of UASs based on mission objectives, client requirements, and application-specific constraints.	-	1	-	-
PC3. Collect, analyze, and document client requirements, operational issues, and mission expectations using structured communication methods.	-	1	-	-
PC4. Read, interpret, and extract relevant technical data, specifications, and instructions from electronic documentation and publications in various formats.	-	1	-	-
PC5. Communicate technical information clearly and effectively through oral, written, and digital means while using appropriate technical terminology and professional language.	-	1	-	-
PC6. Exchange information, instructions, and advice with team members and stakeholders, seek clarifications, and provide constructive feedback using standard communication technologies.	-	1	-	-
PC7. Conduct pre-mission planning meetings with clients and internal teams to confirm mission scope, safety requirements, timelines, and documentation.	-	1	-	-
PC8. Create, update, organize, and maintain design and technical documentation such as schematics, wiring diagrams, bills of material, and knowledge-base or training materials.	-	1	-	-
NOS Total	**-**	**8**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6732
NOS Name	Communication and interpersonal skills (Unmanned Aerial System)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6733: Assemble/repair, and maintain an unmanned aerial system (UAS/RPAS) (Unmanned Aerial System)

Description

This NOS defines the competencies required to assemble, inspect, troubleshoot, repair, and maintain UAS/RPAS platforms and their components to ensure safe, reliable, and efficient flight operations.

Scope

The scope covers the following :

- Assemble/repair, and maintain an unmanned aerial system (UAS/RPAS)

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Select appropriate troubleshooting instructions, tools, equipment, and verified reference documents in accordance with Standard Operating Procedures (SOPs).

PC2. Perform preliminary inspection and functional checks of UASs, following recommended start-up and shutdown practices and observing ESD safety measures.

PC3. Identify malfunctioning electronic components such as resistors, capacitors, diodes, transistors, ICs, motors, GPS modules, cameras, and power systems requiring repair or replacement.

PC4. Disassemble UAS assemblies and components using approved tools and standard work practices, ensuring no damage to functional parts and maintaining ESD compliance.

PC5. Inspect, test, repair, and replace defective electronic and electromechanical components using external power sources, diagnostic tools, and multimeters as per maintenance guidelines.

PC6. Reassemble UASs, perform post-maintenance functional and diagnostic tests, address client feedback where applicable, and prepare maintenance and service reports using prescribed systems and methods.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Comprehensive knowledge of troubleshooting methodologies, Standard Operating Procedures (SOPs), approved maintenance practices, and the selection of appropriate tools, equipment, and verified reference documents.

KU2. Understanding of Unmanned Aerial System (UAS) architecture, including flight controllers, propulsion systems, GPS modules, cameras, power distribution systems, and electronic control circuits.

KU3. Knowledge of the working principles, specifications, and failure symptoms of electronic components such as resistors, capacitors, diodes, transistors, integrated circuits (ICs), motors, and communication modules.

KU4. Understanding of Electrostatic Discharge (ESD) protection measures, safe start-up and shutdown procedures, safe disassembly techniques, and handling precautions to prevent component damage.

KU5. Knowledge of inspection techniques, diagnostic testing using multimeters and external power supplies, repair and replacement guidelines, reassembly procedures, post-maintenance testing, and preparation of service and maintenance reports.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Skill in performing preliminary inspections, start-up and shutdown checks, and functional tests of UASs while adhering to safety and ESD compliance requirements.

GS2. Proficiency in inspecting, testing, repairing, and replacing defective components such as motors, GPS modules, cameras, and power systems using approved tools and diagnostic equipment.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**12**	**-**	**-**
PC1. Select appropriate troubleshooting instructions, tools, equipment, and verified reference documents in accordance with Standard Operating Procedures (SOPs).	-	2	-	-
PC2. Perform preliminary inspection and functional checks of UASs, following recommended start-up and shutdown practices and observing ESD safety measures.	-	2	-	-
PC3. Identify malfunctioning electronic components such as resistors, capacitors, diodes, transistors, ICs, motors, GPS modules, cameras, and power systems requiring repair or replacement.	-	2	-	-
PC4. Disassemble UAS assemblies and components using approved tools and standard work practices, ensuring no damage to functional parts and maintaining ESD compliance.	-	2	-	-
PC5. Inspect, test, repair, and replace defective electronic and electromechanical components using external power sources, diagnostic tools, and multimeters as per maintenance guidelines.	-	2	-	-
PC6. Reassemble UASs, perform post-maintenance functional and diagnostic tests, address client feedback where applicable, and prepare maintenance and service reports using prescribed systems and methods.	-	2	-	-
NOS Total	**-**	**12**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6733
NOS Name	Assemble/repair, and maintain an unmanned aerial system (UAS/RPAS) (Unmanned Aerial System)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6734: Setup, program, and operate the UAS (Unmanned Aerial System)

Description

This NOS defines the competencies required to configure, program, and safely operate UAS platforms, including mission planning, system calibration, flight execution, and compliance with aviation regulations.

Scope

The scope covers the following :

- Setup, program, and operate the UAS

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Conduct diagnostic tests on UAS hardware and software systems to assess operational effectiveness and identify faults in accordance with standard diagnostic procedures.

PC2. Interpret and comply with applicable UAS laws, rules, airspace structures, and Air Traffic Control (ATC) / radio telephony procedures during planning and operations.

PC3. Assemble and disassemble UAS components, sub-systems, and airframes following industry-approved practices and safety requirements.

PC4. Integrate complex sensor systems through mechanical assembly, PCB soldering, and fabrication of wiring harnesses as per technical specifications.

PC5. Perform mechanical and electrical integration of new hardware components and upgrade internal software by installing and validating firmware and software packages.

PC6. Demonstrate UAS functionality to validate flight principles, environmental readiness, and risk mitigation measures, ensuring operational safety and customer satisfaction.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Knowledge of standard diagnostic procedures for assessing UAS hardware and software performance and identifying system faults.

KU2. Understanding of applicable UAS laws, aviation regulations, airspace structures, and ATC/radio telephony procedures for compliant operations.

KU3. Knowledge of industry-approved assembly and disassembly practices for UAS components, sub-systems, and airframes with safety compliance.

KU4. Understanding of mechanical assembly, PCB soldering techniques, wiring harness fabrication, and sensor system integration principles.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**12**	**-**	**-**
PC1. Conduct diagnostic tests on UAS hardware and software systems to assess operational effectiveness and identify faults in accordance with standard diagnostic procedures.	-	2	-	-
PC2. Interpret and comply with applicable UAS laws, rules, airspace structures, and Air Traffic Control (ATC) / radio telephony procedures during planning and operations.	-	2	-	-
PC3. Assemble and disassemble UAS components, sub-systems, and airframes following industry-approved practices and safety requirements.	-	2	-	-
PC4. Integrate complex sensor systems through mechanical assembly, PCB soldering, and fabrication of wiring harnesses as per technical specifications.	-	2	-	-
PC5. Perform mechanical and electrical integration of new hardware components and upgrade internal software by installing and validating firmware and software packages.	-	2	-	-
PC6. Demonstrate UAS functionality to validate flight principles, environmental readiness, and risk mitigation measures, ensuring operational safety and customer satisfaction.	-	2	-	-
NOS Total	**-**	**12**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6734
NOS Name	Setup, program, and operate the UAS (Unmanned Aerial System)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6735: Manual flight demonstration and emergency procedures (Unmanned Aerial System)

Description

This NOS defines the competencies required to perform safe manual flight operations and effectively execute emergency procedures to ensure operational safety and risk mitigation during UAS missions.

Scope

The scope covers the following :

- Manual flight demonstration and emergency procedures

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Collect and document clients' requirements, operational challenges, and mission objectives using structured interaction and requirement-gathering techniques.

PC2. Design and modify UAS platforms and payload configurations, including integration of cameras and other devices, to meet customized client and mission requirements.

PC3. Perform preliminary and pre-flight checks of UASs in accordance with common Standard Operating Procedures to ensure airworthiness and operational readiness.

PC4. Configure radio controllers, communication links, and system parameters for compatibility with newly integrated payloads and devices.

PC5. Pilot UASs to carry out standard flight exercises and mission-specific operations safely, in line with approved flight plans and operational goals.

PC6. Execute abnormal and emergency procedures, including forced landings, completion of emergency checklists, and safe handling of loss of command and control situations.

PC7. Troubleshoot post-crash or abnormal behavior using flight data, logs, simulations, and verified troubleshooting instruction sheets, tools, and equipment.

PC8. Demonstrate UAS functionality to validate performance, apply RPAS/UAS emergency reporting procedures such as flyaway or loss of control, and ensure mission and customer satisfaction.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Knowledge of structured requirement-gathering techniques to document client needs, operational challenges, and mission objectives accurately.

KU2. Understanding of UAS platform design principles, payload integration methods, and configuration of cameras and mission-specific devices.

KU3. Knowledge of Standard Operating Procedures (SOPs) for pre-flight inspections, airworthiness verification, and operational readiness checks.

KU4. Understanding of radio controller configuration, communication link setup, system parameter adjustment, and payload compatibility requirements.

KU5. Knowledge of flight operations, emergency procedures, post-crash diagnostics, flight data analysis, and RPAS/UAS emergency reporting protocols.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to collect, analyze, and document client requirements using structured communication and professional interaction methods.

GS2. Skill in designing, modifying, and integrating UAS platforms and payload systems to meet customized mission requirements.

GS3. Proficiency in conducting pre-flight checks, configuring communication systems, and ensuring operational compatibility of integrated devices.

GS4. Competence in piloting UASs safely for standard and mission-specific flights, including handling abnormal and emergency situations effectively.

GS5. Ability to troubleshoot post-flight issues using logs and diagnostic tools, demonstrate system functionality, and ensure regulatory compliance and customer satisfaction.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**16**	**-**	**-**
PC1. Collect and document clients' requirements, operational challenges, and mission objectives using structured interaction and requirement-gathering techniques.	-	2	-	-
PC2. Design and modify UAS platforms and payload configurations, including integration of cameras and other devices, to meet customized client and mission requirements.	-	2	-	-
PC3. Perform preliminary and pre-flight checks of UASs in accordance with common Standard Operating Procedures to ensure airworthiness and operational readiness.	-	2	-	-
PC4. Configure radio controllers, communication links, and system parameters for compatibility with newly integrated payloads and devices.	-	2	-	-
PC5. Pilot UASs to carry out standard flight exercises and mission-specific operations safely, in line with approved flight plans and operational goals.	-	2	-	-
PC6. Execute abnormal and emergency procedures, including forced landings, completion of emergency checklists, and safe handling of loss of command and control situations.	-	2	-	-
PC7. Troubleshoot post-crash or abnormal behavior using flight data, logs, simulations, and verified troubleshooting instruction sheets, tools, and equipment.	-	2	-	-
PC8. Demonstrate UAS functionality to validate performance, apply RPAS/UAS emergency reporting procedures such as flyaway or loss of control, and ensure mission and customer satisfaction.	-	2	-	-
NOS Total	**-**	**16**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6735
NOS Name	Manual flight demonstration and emergency procedures (Unmanned Aerial System)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6736: Autonomous flight planning and demonstration (Unmanned Aerial System)

Description

This NOS defines the competencies required to plan, configure, and execute autonomous flight missions, including waypoint programming, system validation, and safe operational demonstration in compliance with aviation standards.

Scope

The scope covers the following :

- Autonomous flight planning and demonstration

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Collect, analyze, and document customer requirements, operational challenges, and mission objectives using structured requirement-gathering techniques.

PC2. Conduct pre-mission planning meetings with clients and internal teams to finalize mission scope, timelines, safety considerations, and deliverables.

PC3. Develop detailed flight plans by identifying potential flight hazards, assessing environmental constraints, and coordinating airspace requirements in compliance with regulations.

PC4. Select and apply appropriate troubleshooting instruction sheets, tools, equipment, and verification methods to support UAS software and system validation activities.

PC5. Apply basic programming concepts to develop, modify, and test software components relevant to UAS operations.

PC6. Perform basic software development tasks, including sensor integration, data acquisition, and real-time data processing.

PC7. Configure and operate autopilot systems by integrating hardware and software that process sensor inputs, execute flight plans, and support autonomous decision-making.

PC8. Use autopilot-enabled UAS platforms to perform autonomous operations with minimal human intervention while ensuring safety and system reliability.

PC9. Plan and execute flight test plans to validate new software features, electronics, sensors, and payload integrations, documenting results as per standard practices.

PC10. Use industry-standard software tools and development environments to write, upload, and validate programs for autonomous flight operations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Knowledge of structured requirement-gathering and documentation techniques to accurately capture mission objectives and operational constraints.

KU2. Understanding of pre-mission planning processes, hazard identification, environmental assessment, and regulatory airspace coordination requirements.

KU3. Knowledge of UAS troubleshooting methodologies, verification procedures, and use of instruction sheets, tools, and validation equipment for software and system support.

KU4. Understanding of basic programming concepts, sensor integration principles, data acquisition methods, and real-time data processing in UAS applications.

KU5. Knowledge of autopilot systems, autonomous flight logic, flight test planning procedures, and industry-standard development tools for software validation.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Knowledge of autopilot systems, autonomous flight logic, flight test planning procedures, and industry-standard development tools for software validation.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**20**	**-**	**-**
PC1. Collect, analyze, and document customer requirements, operational challenges, and mission objectives using structured requirement-gathering techniques.	-	2	-	-
PC2. Conduct pre-mission planning meetings with clients and internal teams to finalize mission scope, timelines, safety considerations, and deliverables.	-	2	-	-
PC3. Develop detailed flight plans by identifying potential flight hazards, assessing environmental constraints, and coordinating airspace requirements in compliance with regulations.	-	2	-	-
PC4. Select and apply appropriate troubleshooting instruction sheets, tools, equipment, and verification methods to support UAS software and system validation activities.	-	2	-	-
PC5. Apply basic programming concepts to develop, modify, and test software components relevant to UAS operations.	-	2	-	-
PC6. Perform basic software development tasks, including sensor integration, data acquisition, and real-time data processing.	-	2	-	-
PC7. Configure and operate autopilot systems by integrating hardware and software that process sensor inputs, execute flight plans, and support autonomous decision-making.	-	2	-	-
PC8. Use autopilot-enabled UAS platforms to perform autonomous operations with minimal human intervention while ensuring safety and system reliability.	-	2	-	-
PC9. Plan and execute flight test plans to validate new software features, electronics, sensors, and payload integrations, documenting results as per standard practices.	-	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Use industry-standard software tools and development environments to write, upload, and validate programs for autonomous flight operations.	-	2	-	-
NOS Total	**-**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6736
NOS Name	Autonomous flight planning and demonstration (Unmanned Aerial System)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6737: UAS Advanced piloting and embedded vision (Unmanned Aerial System)

Description

This NOS defines the competencies required to perform advanced UAS piloting maneuvers and utilize embedded vision systems for navigation, object detection, tracking, and intelligent mission execution.

Scope

The scope covers the following :

- UAS Advanced piloting and embedded vision

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate a foundational understanding of photography principles and camera systems used in UAS-based imaging applications.

PC2. Perform preliminary check-ups of UASs to ensure sensor readiness, system integrity, and operational safety prior to data collection missions.

PC3. Integrate advanced sensor systems such as LiDAR, 3D sensing cameras, infrared sensors, and obstacle-avoidance sensors in accordance with mission and application requirements.

PC4. Configure and validate GPS modules and positioning systems to ensure accurate geolocation and alignment of collected data.

PC5. Collect high-quality photography, videography, LiDAR, and infrared data using appropriate mission planning and sensor operation techniques.

PC6. Use software tools to create basic imagery products and preprocess photo, video, LiDAR, and geospatial datasets.

PC7. Implement filtering techniques, including histogram, Kalman, and particle filters, to improve data quality, localization accuracy, and noise reduction.

PC8. Implement Proportional–Integral–Derivative (PID) control methods to support smooth and stable autonomous flight corrections.

PC9. Apply Simultaneous Localisation and Mapping (SLAM) algorithms to enable autonomous navigation and environment mapping.

PC10. Assist in the development and validation of simulation experiences that replicate real-world operating scenarios for testing and training purposes.

PC11. Develop three-dimensional (3D) models and maps using LiDAR and sensor data, and analyze spatial information for actionable insights.

PC12. Use UAS-based imaging, thermography, and sensing techniques to conduct industrial and external asset inspections, including flare stacks, tanks, bridges, and other infrastructure.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Knowledge of photography fundamentals, camera systems, and imaging principles used in UAS-based data acquisition applications.

KU2. Understanding of UAS pre-flight inspection procedures, sensor readiness checks, GPS configuration, and system integrity verification for safe operations.

KU3. Knowledge of advanced sensor technologies such as LiDAR, 3D cameras, infrared sensors, and obstacle-avoidance systems and their integration requirements.

KU4. Understanding of data processing techniques, filtering methods (histogram, Kalman, particle filters), PID control mechanisms, and SLAM algorithms for navigation and mapping.

KU5. Knowledge of geospatial data processing, 3D modeling, simulation validation, and industrial inspection applications using UAS-based imaging and sensing systems.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to conduct pre-mission system checks, configure sensors and GPS modules, and ensure operational safety and data accuracy.

GS2. Skill in integrating and validating advanced sensor systems to meet mission-specific imaging and data collection requirements.

GS3. Proficiency in capturing high-quality photo, video, LiDAR, and infrared data using effective mission planning and sensor operation techniques.

GS4. Competence in preprocessing datasets, applying filtering techniques, implementing PID control and SLAM methods, and generating 3D models and maps.

GS5. Ability to utilize UAS-based imaging and sensing technologies for inspections, simulations, and infrastructure analysis while ensuring professional and safe operations.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**24**	**-**	**-**
PC1. Demonstrate a foundational understanding of photography principles and camera systems used in UAS-based imaging applications.	-	2	-	-
PC2. Perform preliminary check-ups of UASs to ensure sensor readiness, system integrity, and operational safety prior to data collection missions.	-	2	-	-
PC3. Integrate advanced sensor systems such as LiDAR, 3D sensing cameras, infrared sensors, and obstacle-avoidance sensors in accordance with mission and application requirements.	-	2	-	-
PC4. Configure and validate GPS modules and positioning systems to ensure accurate geolocation and alignment of collected data.	-	2	-	-
PC5. Collect high-quality photography, videography, LiDAR, and infrared data using appropriate mission planning and sensor operation techniques.	-	2	-	-
PC6. Use software tools to create basic imagery products and preprocess photo, video, LiDAR, and geospatial datasets.	-	2	-	-
PC7. Implement filtering techniques, including histogram, Kalman, and particle filters, to improve data quality, localization accuracy, and noise reduction.	-	2	-	-
PC8. Implement Proportional–Integral–Derivative (PID) control methods to support smooth and stable autonomous flight corrections.	-	2	-	-
PC9. Apply Simultaneous Localisation and Mapping (SLAM) algorithms to enable autonomous navigation and environment mapping.	-	2	-	-
PC10. Assist in the development and validation of simulation experiences that replicate real-world operating scenarios for testing and training purposes.	-	2	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Develop three-dimensional (3D) models and maps using LiDAR and sensor data, and analyze spatial information for actionable insights.	-	2	-	-
PC12. Use UAS-based imaging, thermography, and sensing techniques to conduct industrial and external asset inspections, including flare stacks, tanks, bridges, and other infrastructure.	-	2	-	-
NOS Total	**-**	**24**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6737
NOS Name	UAS Advanced piloting and embedded vision (Unmanned Aerial System)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training centre based on this criterion.

6. To pass the Qualification Pack, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N6731.Work organization and management (Unmanned Aerial System)	-	8	-	-	8	8
ELE/N6732.Communication and interpersonal skills (Unmanned Aerial System)	-	8	-	-	8	8
ELE/N6733.Assemble/repair, and maintain an unmanned aerial system (UAS/RPAS) (Unmanned Aerial System)	-	12	-	-	12	12
ELE/N6734.Setup, program, and operate the UAS (Unmanned Aerial System)	-	12	-	-	12	12
ELE/N6735.Manual flight demonstration and emergency procedures (Unmanned Aerial System)	-	16	-	-	16	16

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N6736.Autonomous flight planning and demonstration (Unmanned Aerial System)	-	20	-	-	20	20
ELE/N6737.UAS Advanced piloting and embedded vision (Unmanned Aerial System)	-	24	-	-	24	24
Total	**0**	**100**	**-**	**-**	**100**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.